

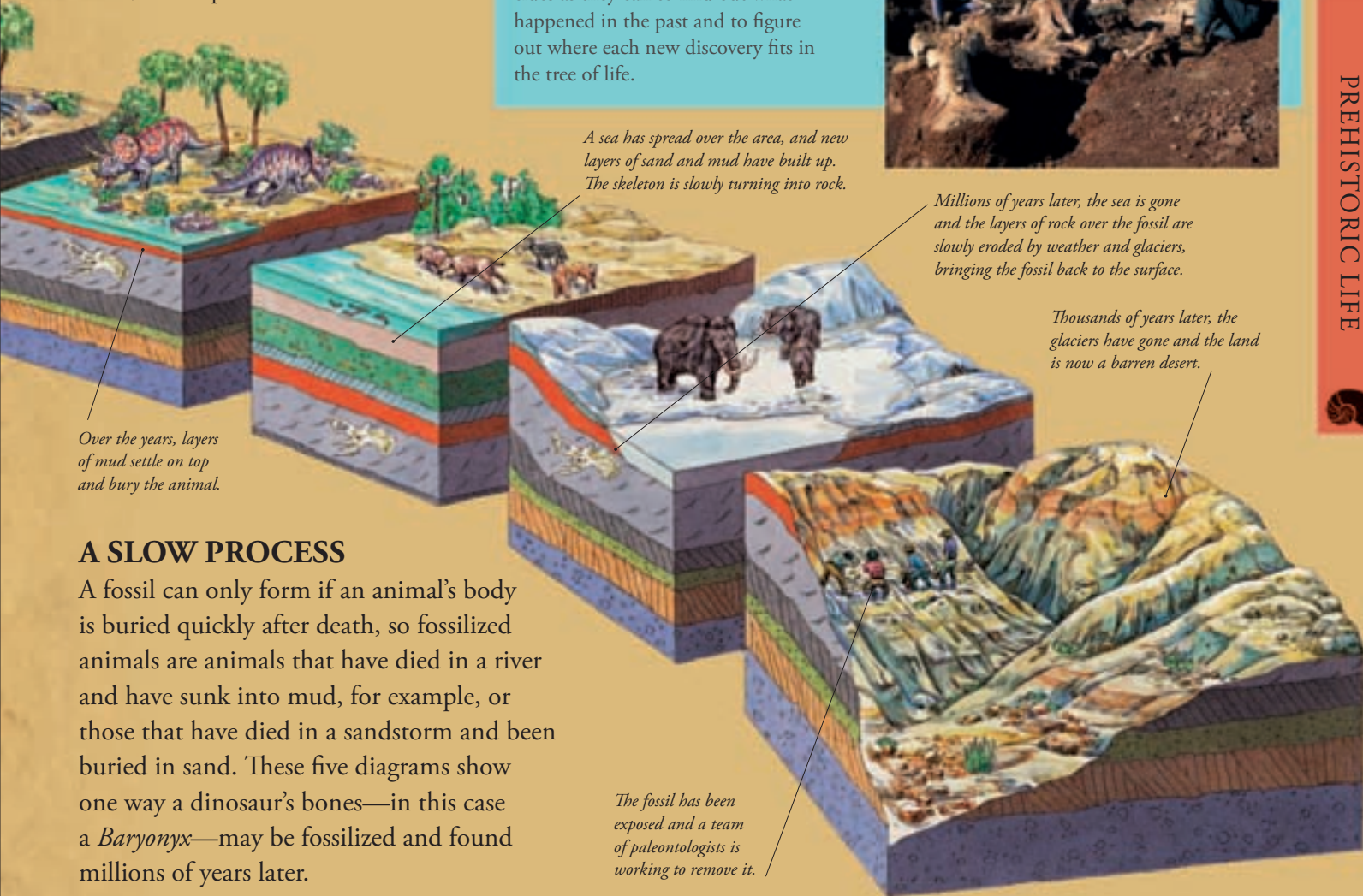
What makes a fossil?

All kinds of living things have been discovered in fossil form. We have unearthed fossilized animal skeletons, skin impressions, footprints, teeth, animal droppings, insects, and plants. The hard parts of an animal, such as the bones, are the parts that fossilize best.



TAKE A LOOK—WHAT IS A PALEONTOLOGIST?

People who study fossils are called paleontologists. Paleontologists may work in the field, digging up new fossils, or in labs or museums. They work like detectives, carefully gathering as many clues as they can to find out what happened in the past and to figure out where each new discovery fits in the tree of life.



Over the years, layers of mud settle on top and bury the animal.

A sea has spread over the area, and new layers of sand and mud have built up. The skeleton is slowly turning into rock.

Millions of years later, the sea is gone and the layers of rock over the fossil are slowly eroded by weather and glaciers, bringing the fossil back to the surface.

Thousands of years later, the glaciers have gone and the land is now a barren desert.

The fossil has been exposed and a team of paleontologists is working to remove it.

A SLOW PROCESS

A fossil can only form if an animal's body is buried quickly after death, so fossilized animals are animals that have died in a river and have sunk into mud, for example, or those that have died in a sandstorm and been buried in sand. These five diagrams show one way a dinosaur's bones—in this case a *Baryonyx*—may be fossilized and found millions of years later.



Petrified Tree trunks, just like bones, can be turned to rock by mineralization over millions of years. Petrified trees still look like logs. Petrification means "change to stone."



External mold Sometimes the original organism dissolves completely, but leaves an impression of itself in the rock. This impression is called a mold.



Natural cast This forms just like an external mold, but the hole then fills in as minerals from water slowly crystallize inside it, forming a rock such as flint.



Trace fossil Occasionally an animal will leave a hint of its presence: a trace. This may be a footprint, a nest, tooth marks, or even droppings. These are called trace fossils.